Welcome to the Workshop

Polyhedra in Lean

Georg Loho Olivia Röhrig Martin Winter

Berlin Mathematics Research Center MATH+

combinatorial synergies

August 24 – September 4, 2026

Welcome

Thanks for coming!

Welcome: about us

Georg Loho

- Professor at FU Berlin
- Local coordination, polytopes

Olivia Röhrig

- PhD student at ZIB
- Lean, technical setup

Martin Winter

- Postdoc at MPI-MiS Leipzig
- polytopes, polyhedral repository in Lean

Further local support

- secretary: Melanie Gessele
- student assistants: Mara Gruß, Valentina Taylor Cerra

Welcome: about the workshop

Setup

- 2 weeks of co-coding and co-learning
- very inhomogeneous group: different career levels, different mathematical and technical skills

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Setup

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Goals of the workshop

- Polyhedral basics in Lean (eventually in mathlib)
  - establish mathlib-community around polyhedral / convex geometry (Zulip activity, PRs, Reviewing)
  - identify the necessary level of abstraction and planning of dependencies
- Foster Lean expertise
- Platform for exchange on recent development around the use of computers in mathematics

Organizational details: rooms

- in Villa (Arnimallee 2): three rooms (basement office, seminar room, ground floor office)
- Pi-Building (Arnimallee 6): SR 031
- Arnimallee 3 (across the street): SR 115, SR 119, SR 120


P
Gemeinsames
Wissenschaftsgebäude
Biodiversität von IGB und FU
Star
120
119
115
A3
Arnimallee
Arnimallee
Arnimallee
Arnimallee
Arnimallee
Villa
A6
031
FB Erziehungswissenschaft
und Psychologie
Haus Hecht

Schedule: online

Organizational details: food & drinks

During work phases

- Snacks and a basis of drinks in the Villa, no explicit coffee breaks
- Kitchen service: please add your name once to the list

Lunch & Dinner

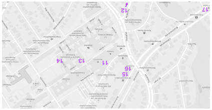
- mostly self-organized, suggestions see homepage

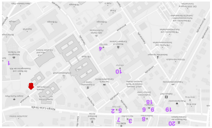
Please sign up for one shift during the workshop. The tasks only take about 5-10 minutes and help keep the shared kitchen running smoothly.
Please also help by taking care of your own dishes throughout the day: after using a cup place it in the dishwasher. Thanks!

	Monday 24.08.	Tuesday 25.08.	Wednesday 26.08.	Thursday 27.08.	Friday 28.08.
Morning (2 people) - Empty the dishwasher - Make coffee	Organizers				
Midday (1 person) - Run the dishwasher if needed - Make coffee					
Afternoon (2 people) - Run the dishwasher - Clean the big coffee pot					

	Monday 31.08.	Tuesday 01.09.	Wednesday 02.09.	Thursday 03.09.	Friday 04.09.
Morning (2 people) (see above)					
Midday (1 person) (see above)					
Afternoon (2 people) (see above)					Organizers

	Place	Walk	Food	Price	Payment	Notes
1	Das Café in der Gartenakademie	7 min	Coffee, pastries and some lunch dishes	10-20€ (lunch)		you can check the menu online; closed on Monday
2	Oliv'chen Kaffeehaus	8 min	Coffee, pastries, sandwiches			
3	Bacis Coffee	8 min	Coffee, pastries, bagels, sandwiches, soup	~ 6€	Card and cash	Best coffee you can find in Dahlem
15	Coffee Cross	15 min	Coffee, pastries, sandwiches and one lunch dish changing every day		Only cash	
4	Eßkultur Berlin (Café)	4 min	Coffee and cakes		Only cash	Weird opening hours
13	Café Kauderwelsch	11 min	Coffee, cake, gözleme		Only cash	Inside Rost- und Silberlaube
14	Pi Café	13 min	Filtercoffee and cake		Only cash	Nice rooftop terrace
11	Mensa FU II	12 min			Only with mensacard*	
12	Mensa Shokudo	24 min	„Japanese" Mensa	More than the usual Mensa	Only with mensacard*	
16	Naya Jouly	15 min	Lebanese Falafel etc.	6-18€	Card and cash	
5	Asia Snack Dahlem	8 min	Wide variety of curries, fried noodles etc.	6-10€	Only cash	
6	Reallygoodlife	9 min	Burgers	~ 10€	Card and cash	
17	Oh my bap	22 min	Korean takeaway	~ 9€		limited options to sit
7	Rewe	7 min	Supermarket		Card and cash	Sometimes there is a Turkish food truck outside selling a variety of tasty spreads.
8	BioBackHaus	9 min	Bakery		Card and cash	
9	Steinecke	9 min	Bakery		Card and cash	
10	Eßkultur (Kantine)	5 min	Changing dishes (check online)	10-20€		





Organizational details: social activities

- **Social evenings**: Wednesday 26.08. and 02.09. coordinated restaurant reservation / game night in Villa
  - Parkcafé Berlin (Fehrbelliner Platz)
  - Ming Dynastie Europa-Center
- **Excursion**: Friday 28.08. afternoon: hike around Krumme Lanke

Informal suggestions for the weekend:

- "Long night of museums" `https://langenachtdermuseen.berlin/en/`
- ...

Questions on logistics?

Group intro

Please raise your hand(s) depending on your level (0,1 or 2 hands):

- You like mathematics.

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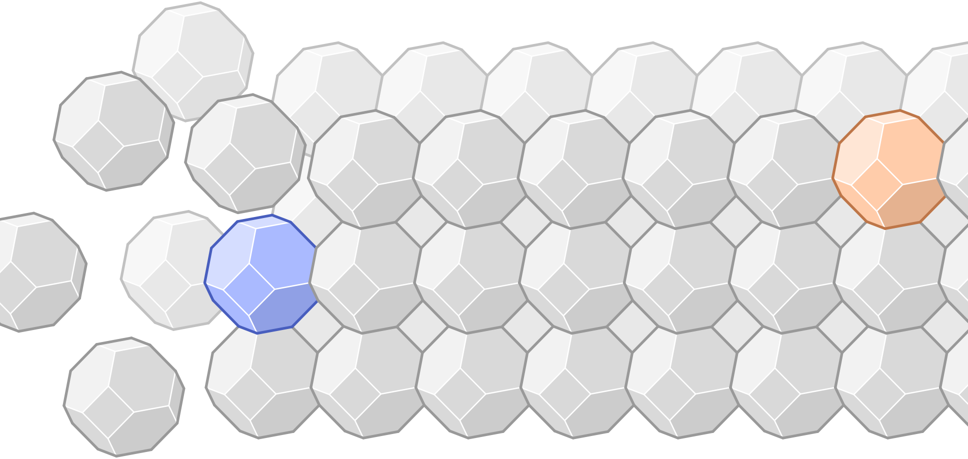
- You know your way around in Berlin.
- You know your way around the campus of FU Berlin

Some opinions

- Why are you here?
- What is your level of expertise?
- What is your goal / your expectation for the workshop?

Thanks for joining us



Polyhedra in Mathlib

What is in mathlib

Foundations:

- linear algebra (+++)
- order structures (posets, lattices, ...) (+++)
- convex cones (++)
- affine spaces (+)

Advanced:

- convex sets, convex hull, ... (Caution: convexity refactor)
- finitely-generated cones
- faces and face lattice for cones
- dual cones

What is <u>not</u> in mathlib

- polytopes and polyhedra (!!)
- H-representations
- Minkowski-Weyl theorem (V $=$ H)
- properties of the face lattice (finite, graded, ...)
- duality theory for polytopes
- edge graphs
- f- and h-vectors
- ...

What is <u>not</u> in mathlib

- polytopes and polyhedra (!!)
- H-representations for cones
- Minkowski-Weyl theorem (V = H) for cones
- properties of the face lattice (finite, graded, ...)
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But some of this is in the polyhedral repository!

Setup for cones

```
import Polyhedral.Mathlib.Geometry.Convex.Cone.Pointed.Basic

variable {K : Type*} [Field K] [LinearOrder K] [IsOrderedRing K]
variable {V : Type*} [AddCommGroup V] [Module K V]

theorem PointedCone.FG.inf (P Q : PointedCone K V) (hP : P.FG) (hQ : Q.FG) :
    FG (P ⊓ Q) := by
  ...
```

Setup for polytopes

```
import Polyhedral.Mathlib.Geometry.Convex.ConvexSpace.Polytope.Basic

open Convexity Affine

variable {K : Type*} [Field K] [LinearOrder K] [IsStrictOrderedRing K]
variable {V : Type*} [AddCommGroup V] [Module K V]
variable {A : Type*} [AffineSpace V A]
variable [ConvexSpace K A] [IsAffineConvexSpace K V A]

theorem IsPolytope.inf (P Q : Set A)
    (hP : IsPolytope K P) (hQ : IsPolytope K Q) :
      IsPolytope K (P ⊓ Q) := by
  ...
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- polytopes live in *affine space* (actually, *convex space*)
- mathlib has affine spaces, but not quite as well developed

Polytopes are V-polytopes

```
def IsPolytope (R) (s : Set X) :=
  ∃ t : Finset X, s = convexHull R t
```

Geometry/Convex/ConexSpace/Polytope/Basic.lean

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Note: `R` is a general *ordered ring*, `X` is a general *convex space* over `R`.

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```
structure Polytope (R X) where
  carrier : Set X
  isPolytope : IsPolytope R carrier
```

Geometry/Convex/ConexSpace/Polytope/Lattice.lean

Generality

Note: per default, our polytopes/cones do not live in $\mathbb{R}^n$!

Cones live in a general module

- over a general semiring
- without an inner product or norm
- without a topology
- and potentially of infinite dimension.

Polytopes live in convex spaces over general semiring.

Overview

linear		affine	
-	$\texttt{SubMulAction}_0$	-	Set
-	Submodule	-	AffineSubspace
-	PointedCone	IsConvexSet	ConexSet
FG	-	IsPolytope	Polytope
IsPolyhedral	PolyhedralCone	IsPolyhedral	Polyhedron

$$\longleftrightarrow$$

(de)homogenization

Cones vs polytopes

Some things are easier to prove on the **cone side**:

- closed under intersection
- dual of polyhedral is polyhedral

Some things are easier to prove on the **polytope side**:

- generated by vertices (atoms)
- face lattice is graded

One transitions back-and-forth between these sides using **(de)homogenization**.

The convexity refactor

`ConvexSpace`

- a space in which you can take convex combinations
- convex sets, polytopes etc are defined in here

Mathlib is in a state of transition from the old convexity API to the new one. Make sure to use the *new* API:

- don't use `IsExtreme`
  $\longrightarrow$ **use** `IsFaceOf`
- don't use `IsExposed`
  $\longrightarrow$ **use** `IsExposedFaceOf`
- don't use `convexHull` from Mathlib.Analysis.Convex.Hull
  $\longrightarrow$ **use** `Convexity.convexHull` from Mathlib.Geometry.Convex.Hull

Faces

Faces

```
structure IsFaceOf (F C : PointedCone R M) : Prop where
  le : F ≤ C
  mem_of_smul_add_mem {x y : M} {a : R} :
    x ∈ C → y ∈ C → 0 < a → a • x + y ∈ F → x ∈ F
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```
theorem IsFaceOf.inf (C F₁ F₂ : PointedCone R M)
    (h₁ : F₁.IsFaceOf C) (h₂ : F₂.IsFaceOf C) :
      (F₁ ⊓ F₂).IsFaceOf C := by
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```
structure Face (C : PointedCone R M) extends PointedCone R M where
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structure Face (C : PointedCone R M) extends PointedCone R M where
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```

`Face C` is also the face lattice of C.

```
def prodOrderIso (C : PointedCone R M) (D : PointedCone R N) :
    Face (C.prod D) ≃o Face C × Face D :=
  ...
```

Contributing

Guiding questions

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I. **Is this really a theorem about polytopes?** Is it maybe about convex sets or maybe even general sets? Does the result hold over general modules and rings with weaker assumptions?

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III. **Is this the best way to prove the result?** Would the proof be easier/shorter if we first prove these five elementary lemmas that are of independent value for mathlib?

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III. **Is this the best way to prove the result?** Would the proof be easier/shorter if we first prove these five elementary lemmas that are of independent value for mathlib?

IV. **Is this result helpful for a broad set of mathematics?** Will it be used as a *tool*, or is it merely an *end*?

Mathlib is a software project

Besides the mathematical content, one needs to consider

- style guides
- naming conventions
- documentation and discoverability
- API design (good general definitions)
- maintainability
- automation (simp, grind, etc.)
- performance (e.g. dependency optimization)
- backwards compatibility

Not everything has to be perfect for your private formalization.

But everything has to be perfect for mathlib!

Other comments

- If you define something new, you now have the responsibility to build sufficient API for the object to interact nicely with the rest of mathlib.
- If you prove something, ask yourself, which nearby statement are also true and equally important/useful. Implement those as well (e.g. sup/inf, union/inter)
- Don't just translate a standard math book into Lean. Plan ahead carefully.
- This workshop is experimental. There is only so much we can achieve in these two weeks. We are hoping to motivate many of you to join the mathlib project, perhaps for polyhedral theory, perhaps more generally

LLM usage

- LLMs can be useful for:
  - syntax
  - searching mathlib
  - understanding the codebase
  - stuff you don't understand
- LLMs are probably harmful for:
  - coding for beginners
  - definitions and theorem statements
  - writing long proofs

Essentially no AI written code makes it into mathlib!

GIT

- **workshop repository:** all files, all history. fork off the `Polyhedral` repo
- **clone:** get a local copy of the repository
- **main branch:** in sync with the official repo
- **branch:** mark the moment you start changing things. make a branch when
  - starting a new project with your group
  - trying things your group might need later (sub-branch off your group branch!)
  - implementing a general feature other groups might need
- **merge:** integrate other people's changes by merging feature branches into yours
- **pull request:** start review process to get your changes into official repo

- the workshop repository provides basic functionality for polyhedral geometry/combinatorics, **but you cannot edit it!**
- you cannot push on main
- each group creates a branch on which they can work freely
- create sub-branches as you need
- if group A needs code from other group B:
  - group B splits off feature on new branch
  - group A merges this new branch

Group work

Group work

Groups: ideally 5 – 6 people, ≥ 1 knowledgeable about each of

- the mathematics of the project
- Lean/mathlib
- git (and other parts of the tech setup)
- the polyhedral repo
- the FU campus

Once in groups ...

- introduce yourself to each other.
- make a plan, discuss the goal and structure of the project.
- distribute and parallelize the work.
- **make a branch** on the fork for your *group*.
- **make a thread** in the Workshop Zulip for each *project*.
- if you need something from other groups: ask in the project's Zulip thread.
- make `sorry`-s in order to stay independent.

Gemeinsames
Wissenschaftsgebäude
Biodiversität von IGB und FU
P
Star
120
119
115
A3
Arnimallee
Arnimallee
Arnimallee
Arnimallee
Arnimallee
Villa
FB Erziehungswissenschaft
und Psychologie
A6
031
Haus Hecht

Group formation

Topic

Expertise

Crucial

Polyhedra ◯ Lean ◯ git ◯

Optional

Berlin ◯ Polyhedra Repository ◯

Projects

Project ideas

- H-representations
- polytope intersections
- polar duality
- small and low-dimensional polytopes
- simple/simplicial polytopes
- tangent cones and recession cones
- Minkowski sums
- edge graphs
- abstract polytopes
- combinatorial identities
- lattice polytopes
- shellings
- matroid polytopes
- abstract convexity
- polytopes in non-Euclidean spaces
- tropical polytopes
- triangulations
- polyhedral complexes
- polyhedral fans
- symmetry groups
- special polytopes
- support functions
- Gale duality
- ...

You don't have to aim high ...

Build a concept:

- dimension of a polytope
- edge graph
- simplex
- tangent cone
- relative interior

Prove basic properties: e.g. face lattices are ...

- bounded
- graded
- diamond property
- flag connected
- complemented

Other fundamental results:

- characterize polytopes of dimension 0, 1 and 2.
- every polytope is the projection of a simplex.

Sample project I: edge graphs

- define the edge graph $G_P = (V, E)$.

Things to aim for:

- G_P is connected
- $|V| \geq \dim(P)$
- Balinski's theorem: G_P is d-connected

Requirement:

- tangent cones
- graded face lattices
- small polytopes

Sample project II: f- and h-vectors

- define the f- and h-vector of a polytope

Things to aim for:

- Euler-Poincaré identity: $f_{-1} - f_0 + f_1 - \cdots \pm f_d = 0$
- Dehn-Sommerville relations: $h_d = d_{d-i-1}$.

Requirement:

- dimensions of polytopes
- graded face lattices

Project suggestions?

Project selection